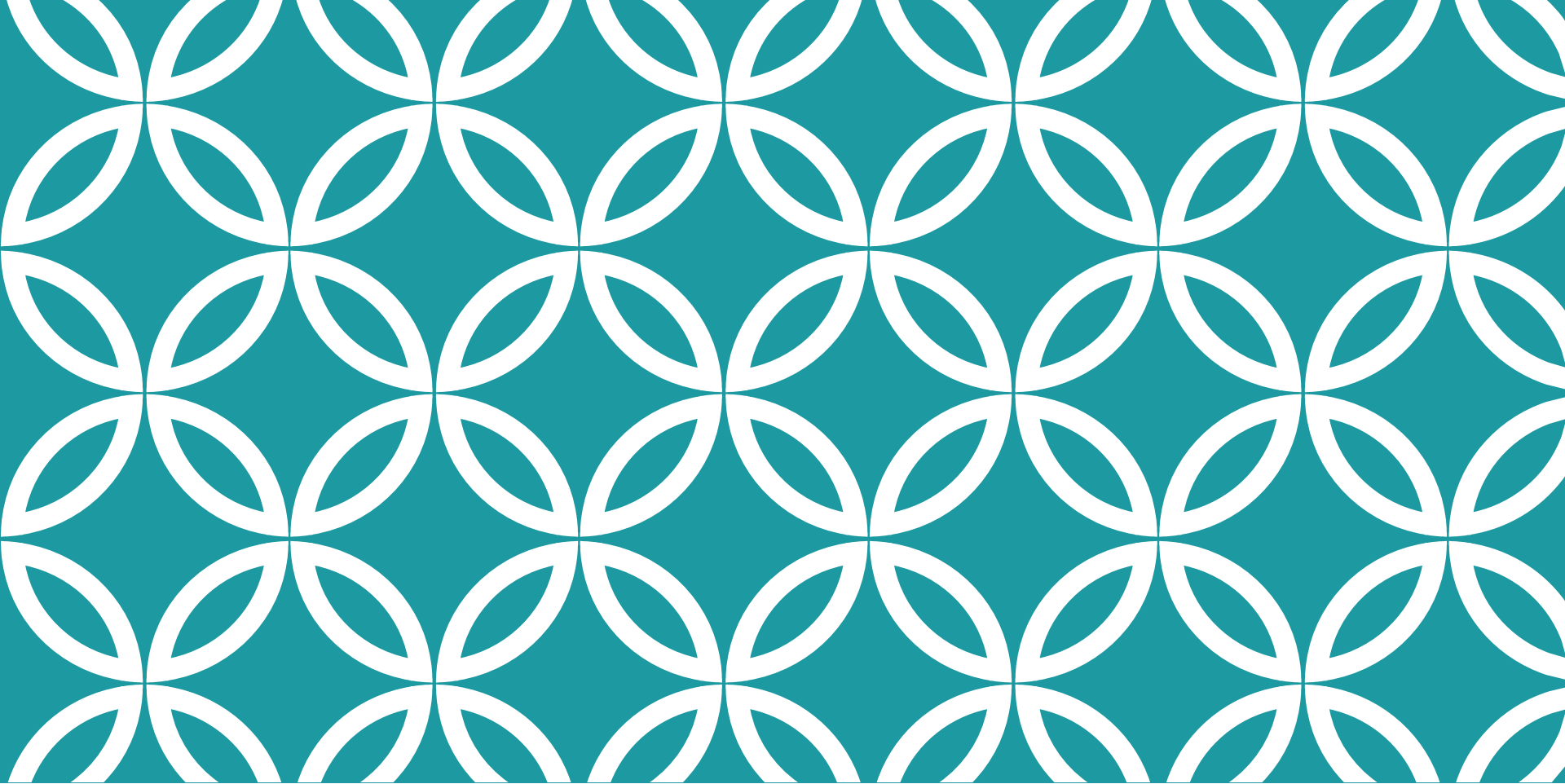


SPEECH SIGNAL REPRESENTATION

Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.
3. Digital Signal Processing John G. Proakis Dimitris K. Manolakis



REVIEW OF DIGITAL SIGNAL PROCESSING

Basic Concepts
Continuous vs Discrete

BASIC CONCEPTS

“A *signal* is defined as any physical quantity that varies with time, space, or any other independent variable or variables.”

A time-series signal is a function of time. e.g. $s_1(t) = 10t$, $s_2(t) = 5t^2$

The simplest periodic signal is a sine wave and can be represented as

$$x(t) = A \sin(2\pi f t + \phi)$$

where

$x(t)$ = instantaneous value (signal amplitude) at time t

A = amplitude (maximum value of the wave),

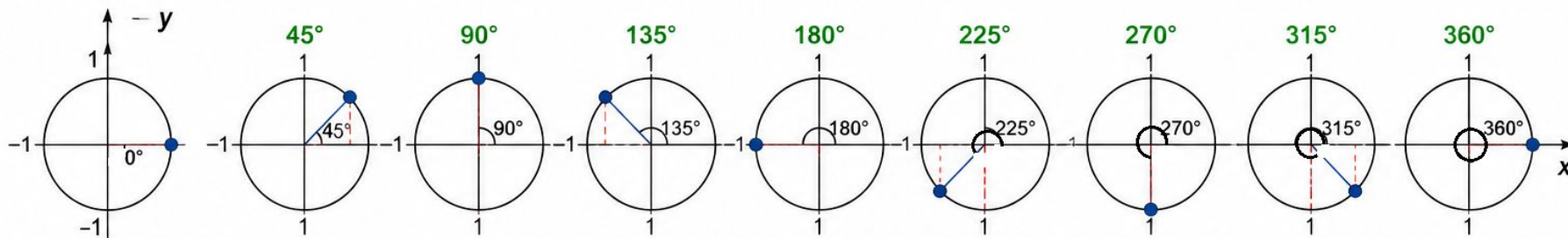
f = frequency (Hz),

t = time (seconds),

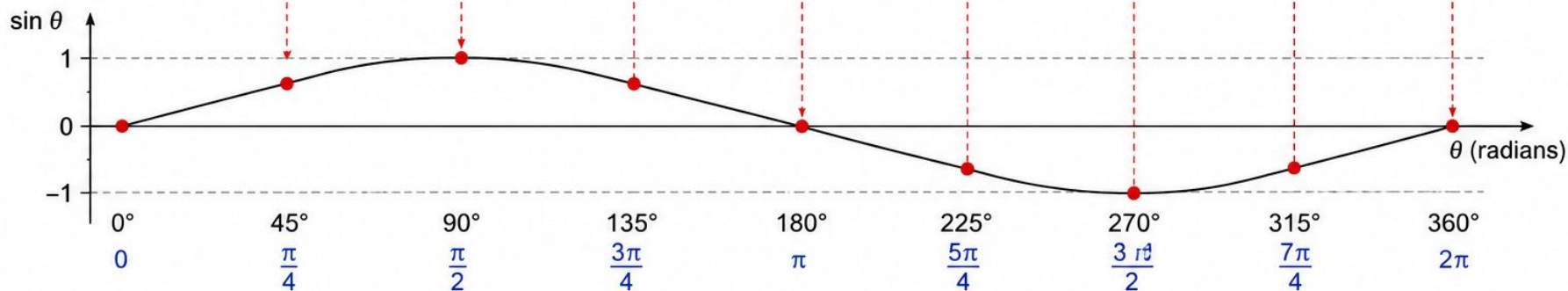
ϕ = phase angle (radians)

BASIC CONCEPTS

1. Motion on the Unit Circle



2. Vertical Projection ($y = \sin \theta$)



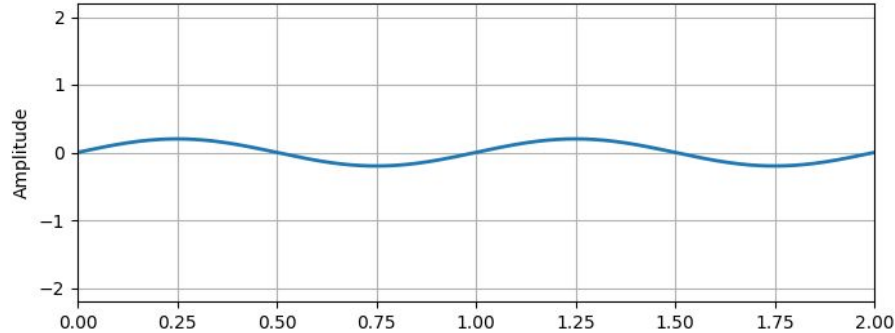
3. Mapping Table

θ (degrees)	0°	45°	90°	135°	180°	225°	270°	315°	360°
θ (radians)	0	$\pi/4$	$\pi/2$	$3\pi/4$	π	$5\pi/4$	$3\pi/2$	$7\pi/4$	2π
$\sin \theta$ (y)	0	$\sqrt{2}/2$	1	$\sqrt{2}/2$	0	$-\sqrt{2}/2$	-1	$-\sqrt{2}/2$	0

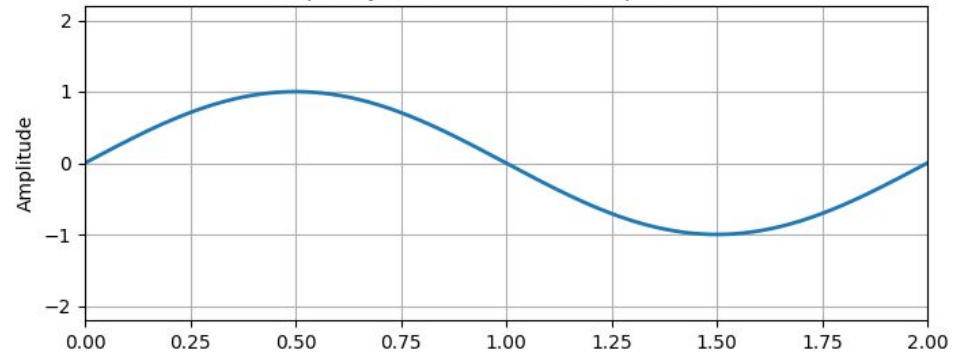
As a point P moves counterclockwise on the unit circle with angle θ , its vertical projection on the y -axis is $\sin \theta$.
Plotting this projection versus θ gives the sine wave.

BASIC CONCEPTS

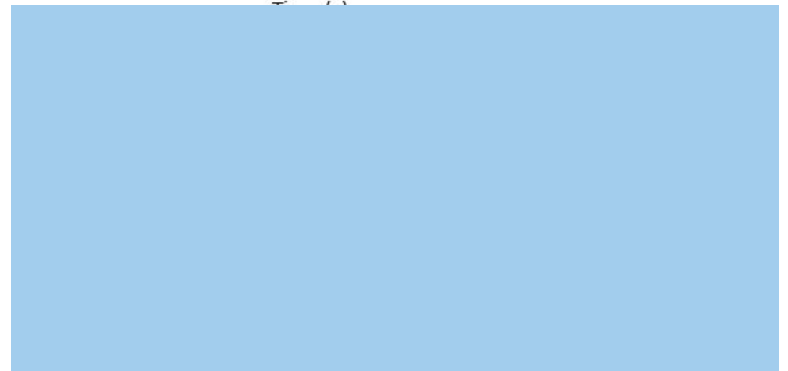
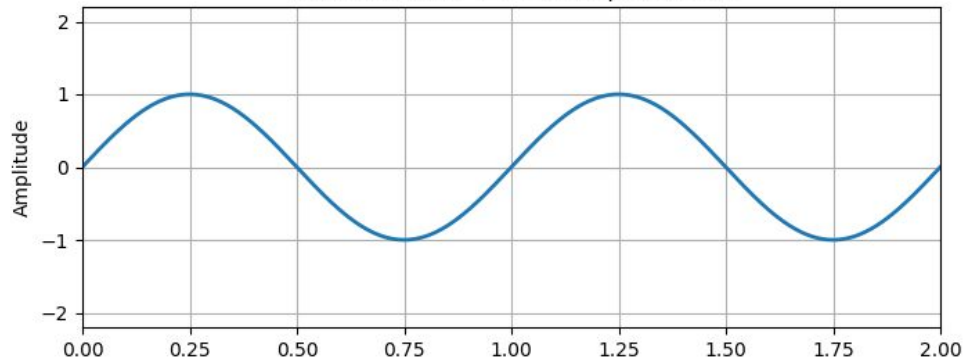
Amplitude: $A=0.20$, $f=1.00$ Hz, $\phi=0.00$ rad



Frequency: $A=1.00$, $f=0.50$ Hz, $\phi=0.00$ rad



Phase: $A=1.00$, $f=1.00$ Hz, $\phi=0.00$ rad



BASIC CONCEPTS

Any natural (e.g. speech) signal can be approximated to arbitrary accuracy as **a sum of sinusoidal components having different frequencies, amplitudes, and phases.**

$$x(t) = \sum_{k=1}^{\infty} A_k \sin(2\pi f_k t + \phi_k).$$

where

A_k = amplitude,

f_k = frequency,

ϕ_k = phase.

Detailed discussion in *Fourier Transform*.

CONTINUOUS-TIME VS DISCRETE-TIME SIGNALS

- A **continuous-time signal** is a signal that is defined for **every instant of time** within a given interval.

When defining continuous sinusoidal signal $x(t)$, t is a continuous variable

$$x(t) = A \sin(\omega_c t + \phi), \quad -\infty < t < \infty$$
$$\omega_c = 2\pi f \text{ (rad/s)}$$

- A **discrete-time signal** is a signal whose value is defined only at discrete instants of time. It is represented as $x[n]$, where n is an integer index.

A discrete-time sinusoidal signal may be expressed as

$$x[n] = A \sin(\omega n + \phi), \quad n = 0, \pm 1, \pm 2, \dots$$

ω = **digital angular frequency**(rad/sample)

n = sample index, n can have infinite range $-\infty < n < \infty$

CONTINUOUS-TIME VS DISCRETE-TIME SIGNALS

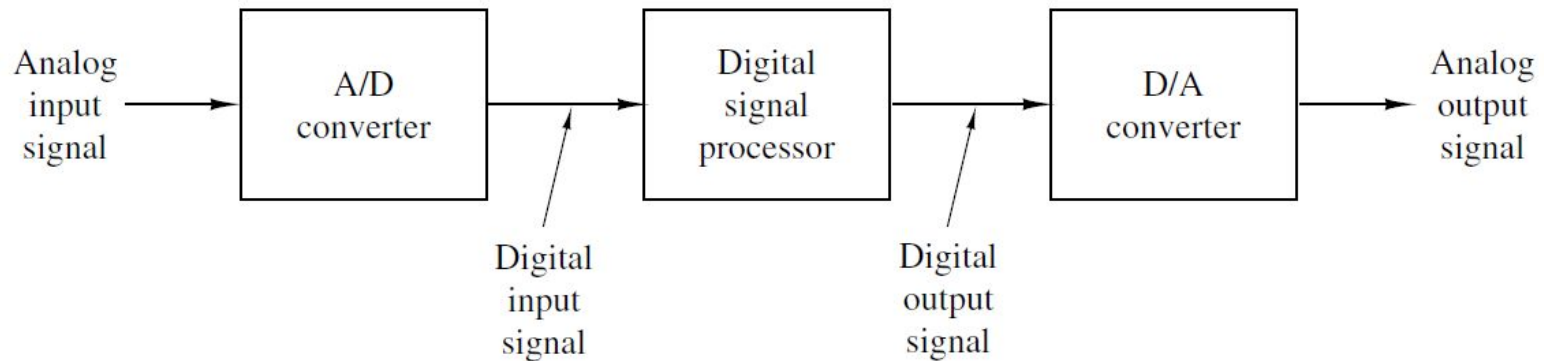
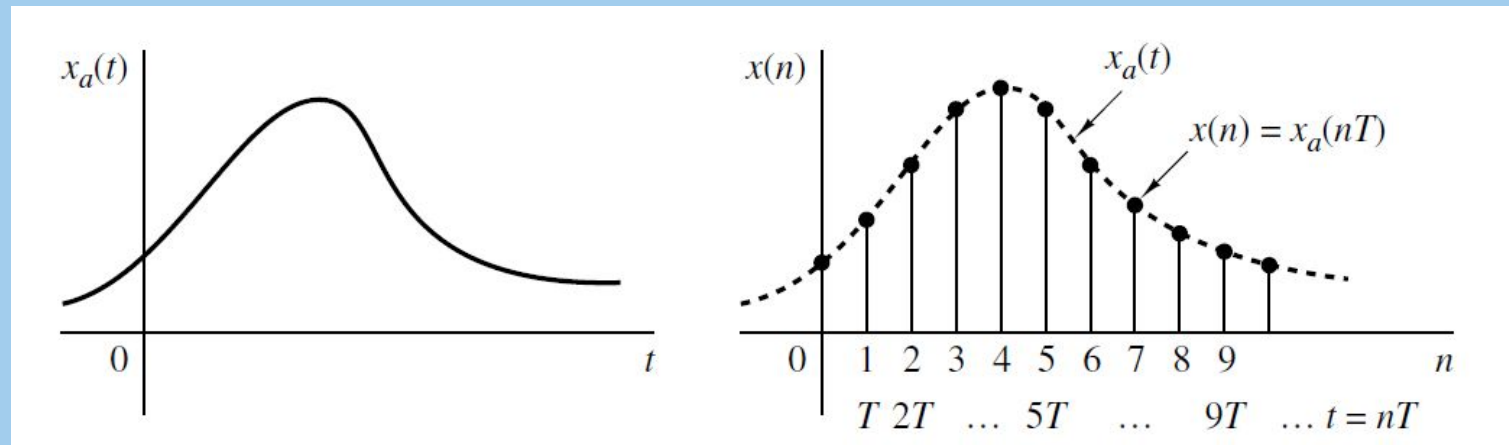
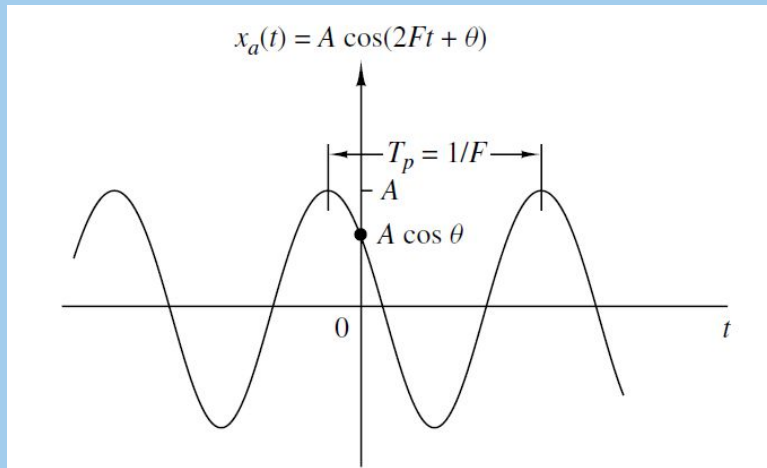


Figure 1.3 Block diagram of a digital signal processing system.

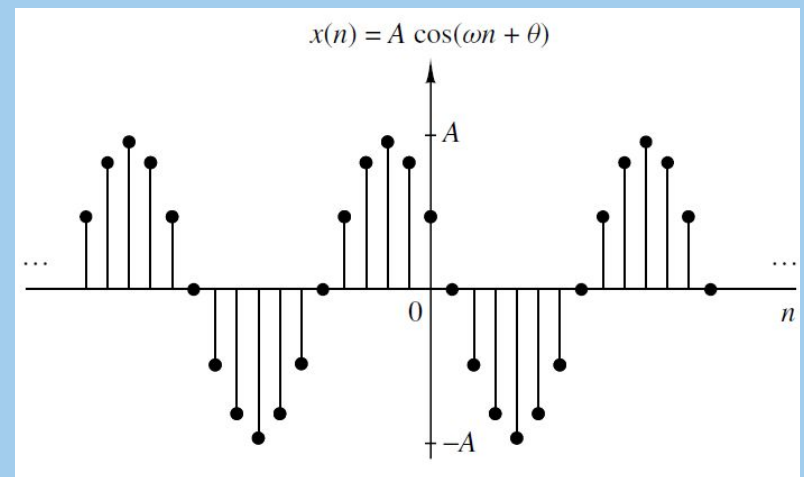


Sampling an analog signal to a digital signal

CONTINUOUS-TIME VS DISCRETE-TIME SIGNALS



Example of an analog sinusoidal signal.



Example of a discrete-time sinusoidal signal ($\omega = \pi/6$, and $\theta = \pi/3$).

Continuous-Time Signal

Represented as $x(t)$

Defined for every instant of time

Time is a continuous variable

Example: Human speech before recording, Voltage across an electrical circuit

Discrete-Time Signal

Represented as $x[n]$

Defined only at discrete sampling instants

Time is represented by an integer sample index

Example: Recording the value of gold every day, Recorded speech sampled at 16 kHz

CONTINUOUS-VALUED VS DISCRETE-VALUED SIGNALS

If a signal takes on all possible values on a finite or an infinite range, it is said to be a continuous-valued signal.

Alternatively, if the signal takes on values from **a finite set of possible values**, it is said to be a discrete-valued signal.

A **discrete-time signal having a set of discrete values** is called a *digital signal*.

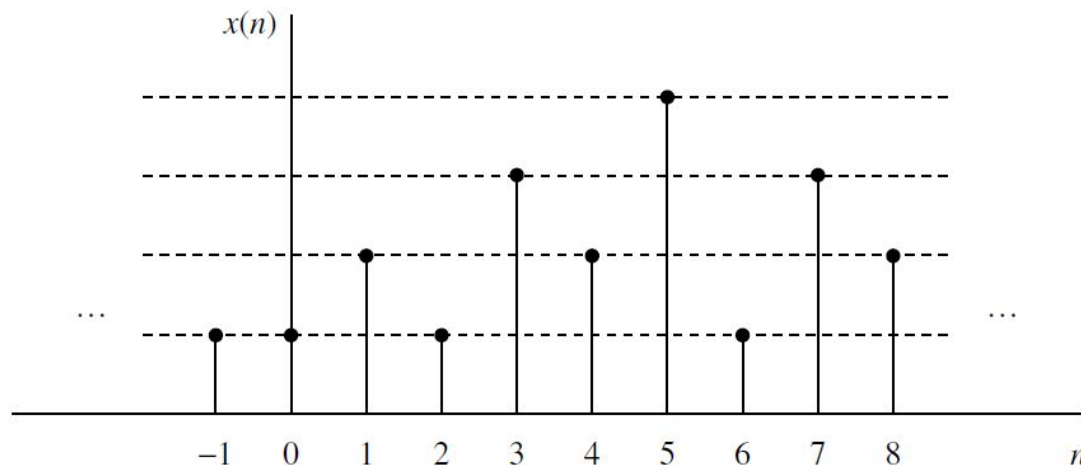
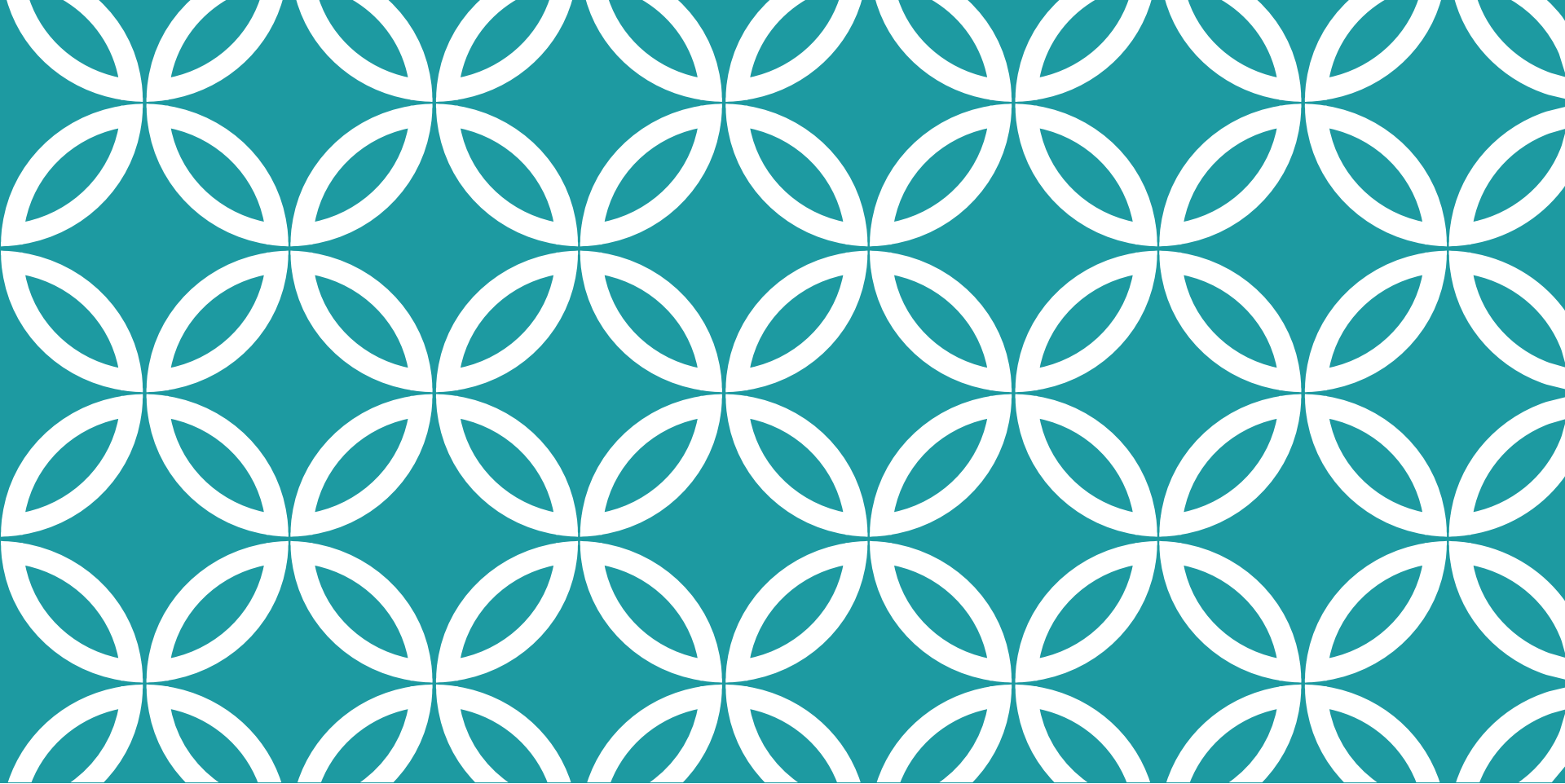


Figure 2.5 Digital signal with four different amplitude values.



DIGITIZATION OF SPEECH

Speech Signal
Discretization
Quantization

SPEECH SIGNAL

- Speech is an acoustic wave. A source vibrates (e.g. vocal cords, speaker). The vibration causes nearby particles to move back and forth, creating regions of compression and expansion. The disturbance in the particle travels as acoustic wave.
- It is a **non-stationary signal**, meaning its statistical properties change over time.
- It is **quasi-periodic** in nature, with voiced sounds showing periodicity due to vocal cord vibrations, while unvoiced sounds resemble noise.

Characteristics of Acoustic Waves

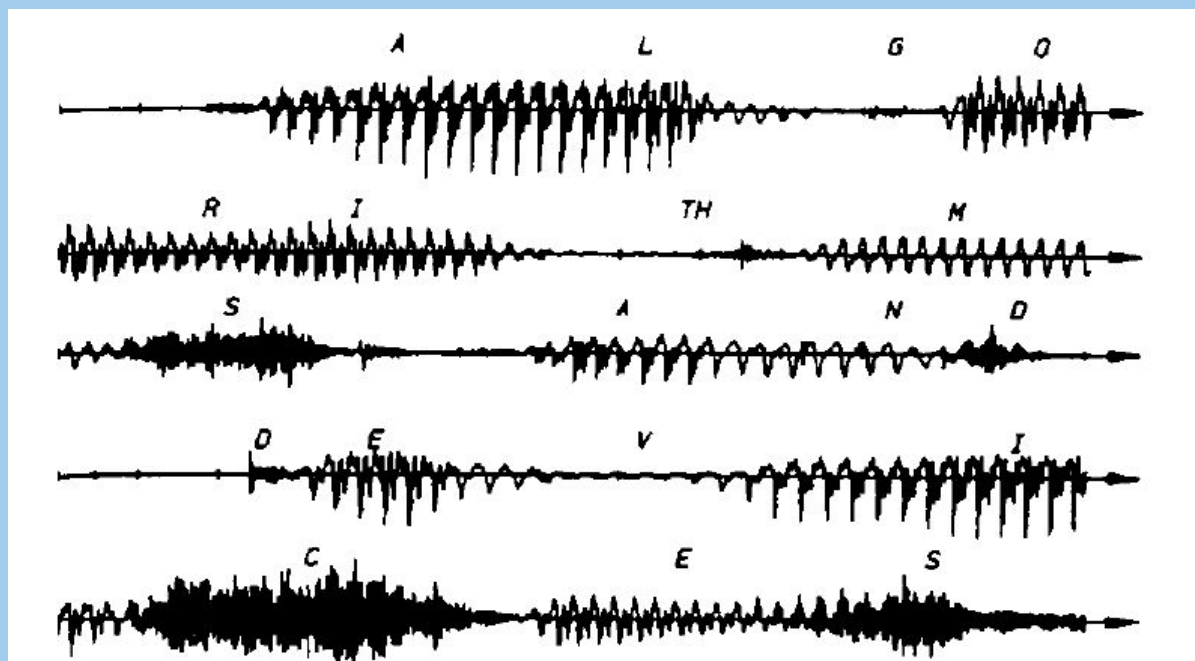
Frequency(f): determines the pitch of the sound (Hz)

Amplitude(a): determines the loudness of the sound.

Wavelength (λ): distance between successive compressions.

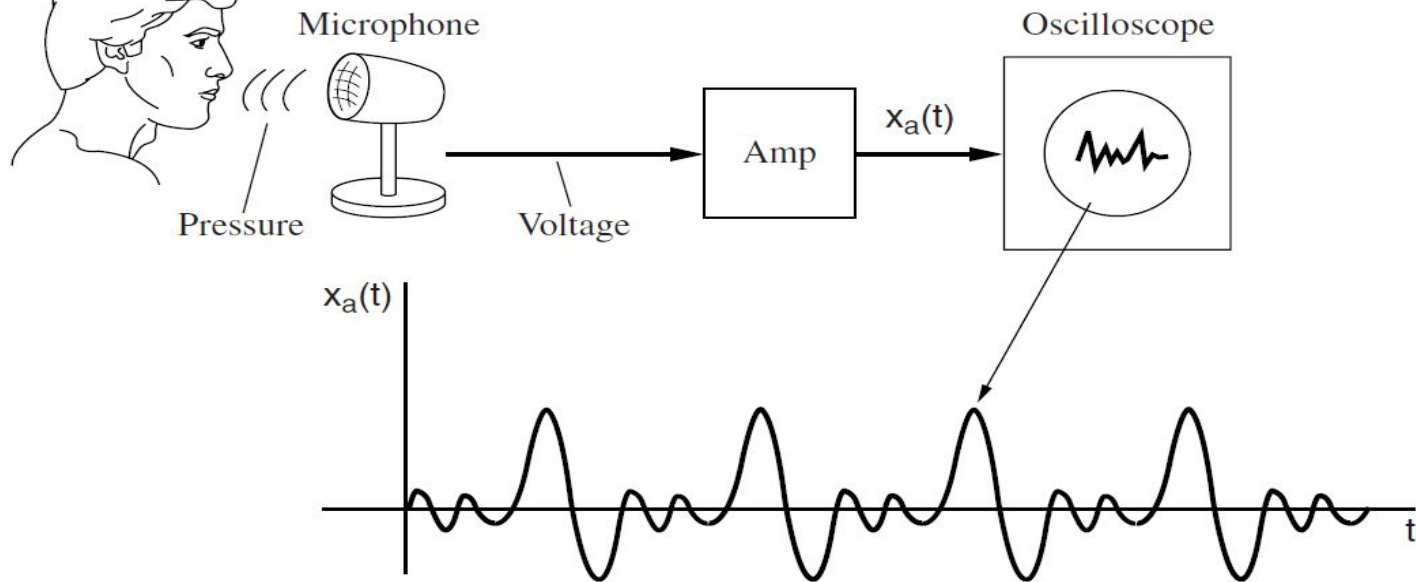
SPEECH SIGNAL

- The basic linguistic units — phonemes, syllables, and words — are embedded in continuous acoustic streams.
- Variability arises from speaker differences (age, gender, accent), speaking rate, and background noise. This variability makes speech representation challenging but also rich in information.



DIGITIZATION

Measurement of an analog speech waveform



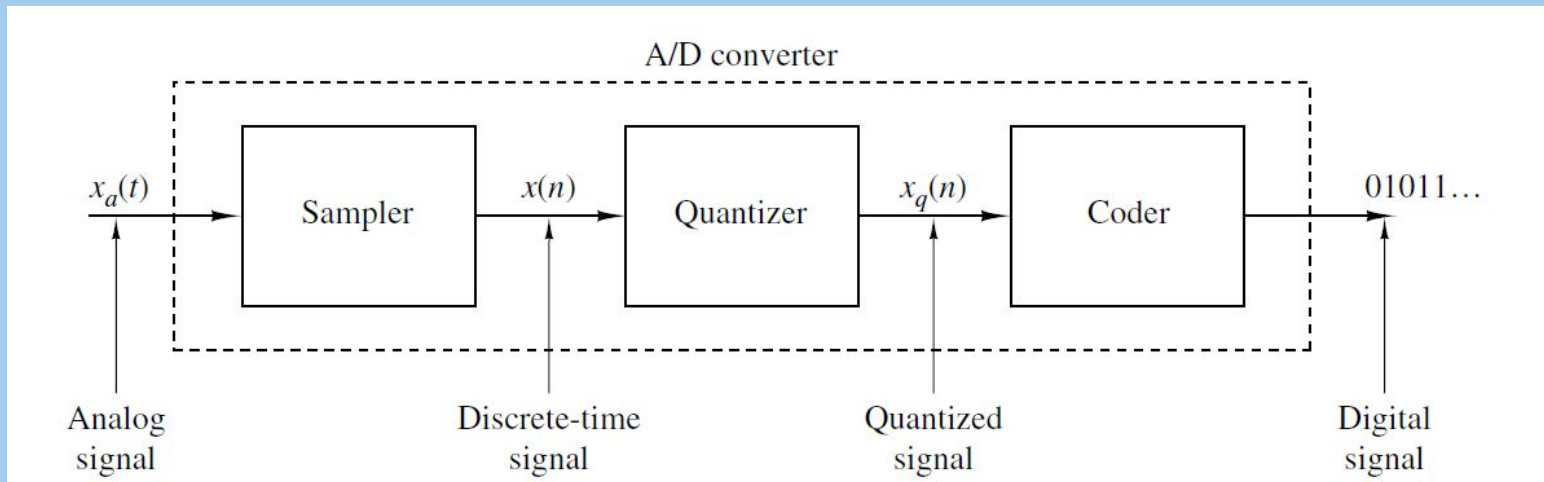
- The pressure wave from an acoustic can be transduced to an electrical signal with a microphone.
- The microphone voltage can be amplified and viewed on an oscilloscope.
- The resulting analog waveform is denoted by $x_a(t)$, a *continuous-time signal* (an analog waveform).

DIGITIZATION

- For digital speech processing input, acoustic signal is converted to electrical signal (microphone).
- For digital speech processing output, electrical signal is to be converted to acoustic signal (speaker).
- A digital programmable system allows flexibility in reconfiguring the digital signal processing operations simply by changing the program.
- Reconfiguration of an analog system usually implies a redesign of the hardware followed by testing and verification to see that it operates properly.

DIGITIZATION

- The process to convert analog signals into digital form is called *analog-to-digital (A/D) conversion*, and the corresponding devices are called *A/D converters (ADCs)*.
- *Sampling* - Taking “samples” of the continuous-time signal at discrete time instants.
- *Quantization* - Conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal.
- *Coding* - In the coding process, each discrete value $x_q(n)$ is represented by a b -bit binary sequence



DISCRETIZATION

- $x_a(t)$ is sampled at uniformly spaced time instants with T being the sampling interval.
- $1/T$ is called as sampling frequency (F_s).
- Output of the sampler (a *continuous-to-discrete (C/D) converter*) is a series of numbers $x_a(nT)$ whose representation is simplified as
$$x[n] = x_a(nT)$$
- The series $x[n]$ is referred to as a discrete-time signal or sequence.

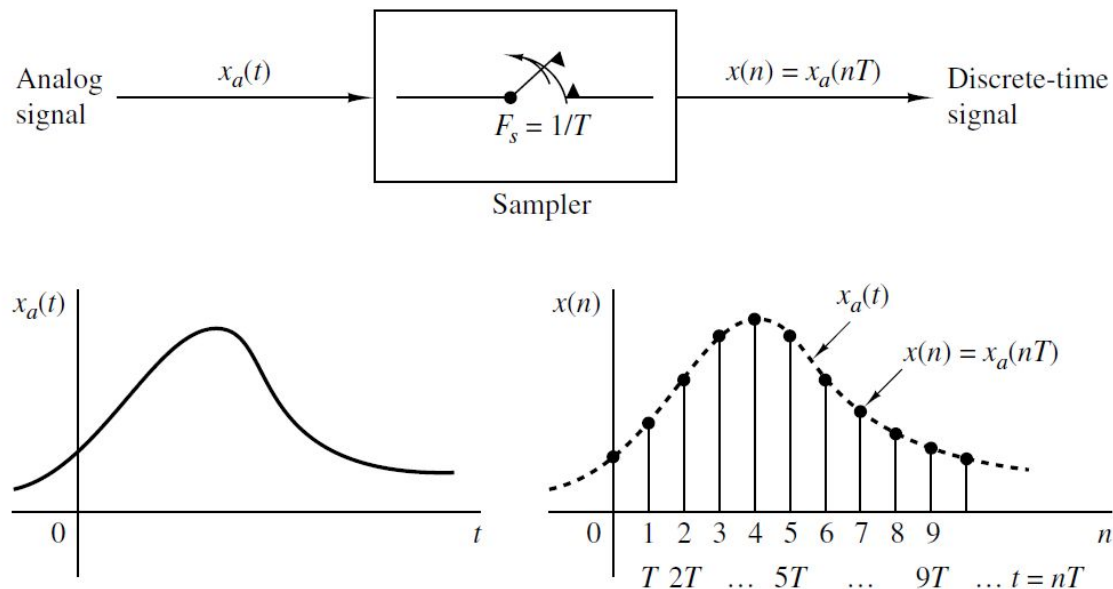


Figure 4.3 Periodic sampling of an analog signal.

NYQUIST-SHANNON SAMPLING THEOREM

The Sampling theorem states that to perfectly reconstruct a continuous-time signal from its samples without distortion, the sampling rate must be at least *twice the highest frequency* present in the signal.

$$\text{Sampling rate } (f_s) \geq 2 \times \text{maximum signal frequency } (f_{max})$$

This minimum rate is called the *Nyquist rate* ($2f_{max}$), and half of the sampling frequency is the *Nyquist frequency* ($f_s/2$).

- Human hearing extends up to ~ 20 kHz. CDs use **44.1 kHz sampling rate**, slightly above 2×20 kHz, to ensure faithful reproduction.
- Telephony systems often use **8 kHz sampling rate** (Nyquist frequency = 4 kHz), which is enough for intelligible voice transmission.

If sampling frequency is too low, then high-frequency details are missed and **aliasing** occurs.

QUANTIZATION

- Quantization is the process to **convert continuous amplitudes into discrete levels** so signals can be encoded in binary.
- The purpose of quantization is to make a continuous-amplitude signal representable in the digital domain by restricting it to a finite set of discrete values.
- Without quantization, sampled signals would have infinitely many possible amplitudes, making them impossible to store or process digitally.
- It reduces the amount of information needed to represent a signal, enabling efficient storage and transmission.
- It is done after sampling, during Analog-to-Digital Conversion (ADC), and introduces a small error called **quantization noise/error**.

BIT RATE AND STORAGE

Sampling converts a continuous-time signal into discrete-time samples.

Quantization converts each sample's amplitude into one of a limited number of allowed values.

Encoding assigns binary codes to these quantized levels for digital representation.

Based on the sampling and quantization, storage size will be affected.

$$\text{File Size (bits)} = f_s \times N \times T, \text{ bit rate (bps)} = f_s \times N$$

where

f_s = **sampling frequency** (samples per second)

N = **bits per sample** (quantization resolution)

T = **duration of recording** (seconds)

To convert to bytes:

$$\text{File Size (bytes)} = \frac{f_s \times N \times T}{8}$$

EXAMPLE EXERCISE

A speech signal is uniformly quantized using **8 bits per sample** over a range of **± 2 V**. Find the **quantization step size (Δ)**. If the sampling frequency 10 Hz, then what will be **the size of the file for a 10 sec recording?**

Solution: Number of levels $L = 2^N = 2^8 = 256$

Range = 4 V (from -2 V to +2 V)

$$\Delta = \frac{\text{Range}}{L} = \frac{4}{256} = 0.015625 \text{ V}$$

So, Step size = **0.0156 V**

Sampling frequency = 10 Hz $\rightarrow$ 10 samples per second

Each sample = 8 bits = 1 byte

Data per second = $10 \times 1 \text{ byte} = 10 \text{ bytes}$

For 10 seconds,

File size = $10 \times 10 = 100 \text{ bytes}$

CLASS EXERCISE

A speech signal is uniformly quantized using 16 bits per sample over a range of ± 5 V. Find the quantization step size (Δ). If the sampling frequency 12 Hz, then what will be the size of the file for a 10 sec recording?

QUANTIZATION ERROR

Quantization error is the **difference between the original sample value and its quantized (rounded) value.**

$$e[n] = x[n] - x_q[n]$$

where $x[n]$ = original sampled signal

$x_q[n]$ = quantized signal

$e[n]$ = quantization error (or quantization noise)

Original ($x[n]$)		Error ($e[n]$)
2.3	2	0.3
3.8	4	-0.2
5.1	5	0.1
6.6	7	-0.4

QUANTIZATION ERROR

Maximum Quantization Error –

For a **uniform rounding quantizer**, the error always lies between

$$-\frac{\Delta}{2} \leq e[n] \leq \frac{\Delta}{2}, \text{ where } \Delta \text{ is the } \mathbf{quantization\ step\ size}.$$

Mean squared Quantization Error –

If the quantization error is assumed to be uniformly distributed, the average quantization noise power is

$$\sigma_e^2 = \frac{\Delta^2}{12}$$

Exercise:

Suppose there is a 8-bit ADC with input range: -1V to $+1\text{V}$. Find the Quantization Step size, Maximum Quantization Error and Mean squared Quantization Error.